IGCSE Sets

What is a set?

In maths, it's often useful to represent **a collection of items**.
We use curly braces to indicate a **set** of items...

$$\{-4, 1, 3\}$$



A set is a collection of items with 2 properties:

a. ?

b. ?

What is a set?

Is it a set?





$\{-3.5, 2, 9\}$ False True

$\{4, 5, 5, 6\}$ False True

$\{1\}$ False True

A set with one thing in it is known as a **singleton**.

$\{\{1,2\}, \{3,4\}\}$ False True

It's possible to have a set of sets!

$\{red, blue, green\}$ False True

Sets need not consist of numbers.

Are these sets the same?

$\{3,1,2\} = \{1,2,3\}$ False True

Finite Sets vs Infinite Sets

$\{-4, 1, 3\}$

The examples with seen have been **finitely** large sets.

But it's also possible to have sets which are **infinitely** large…

"the set of all positive integers (whole numbers)"

"the set of all odd numbers"

(At A Level and beyond, the symbol $\mathbb{N}$ is used for such a set)

Far far beyond the syllabus:
We could construct such a set using

$$\{2k + 1 \mid k \in \mathbb{Z}\}$$

which means "all possible numbers of the form $2k + 1$, where k is any integer"

The Empty Set

We can also have a set with nothing in it!
(think of it as an empty bag)

It is known as the **empty set**:



We'll see why it's useful next lesson.

Venn Diagrams

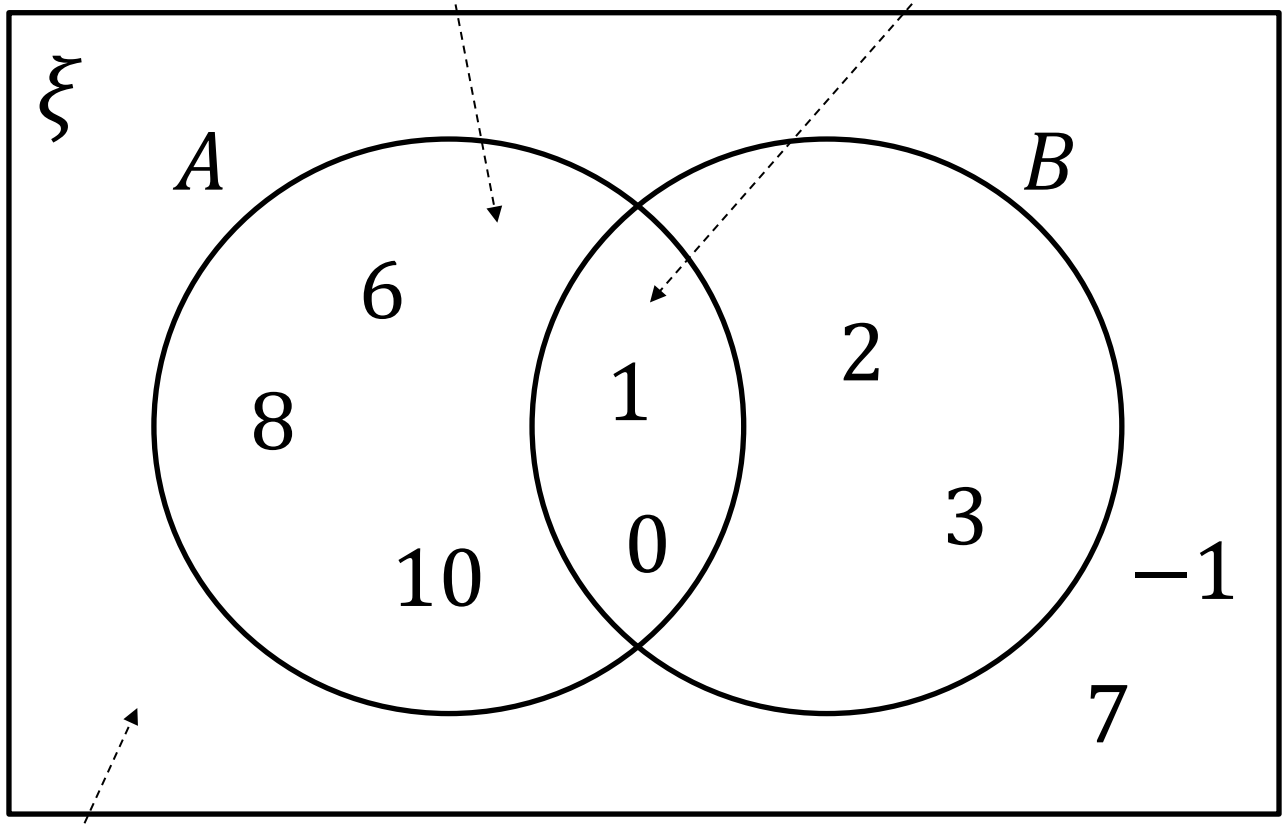
Venn Diagrams are a way of showing the items in each set.

What does this region represent?

?

What does this region represent?

?



What does this region represent?

?

Why the rectangular box?

?

Example

ξ = whole numbers from 1 to 15

A = set of all prime numbers

B = set of all numbers one less than a power of 2

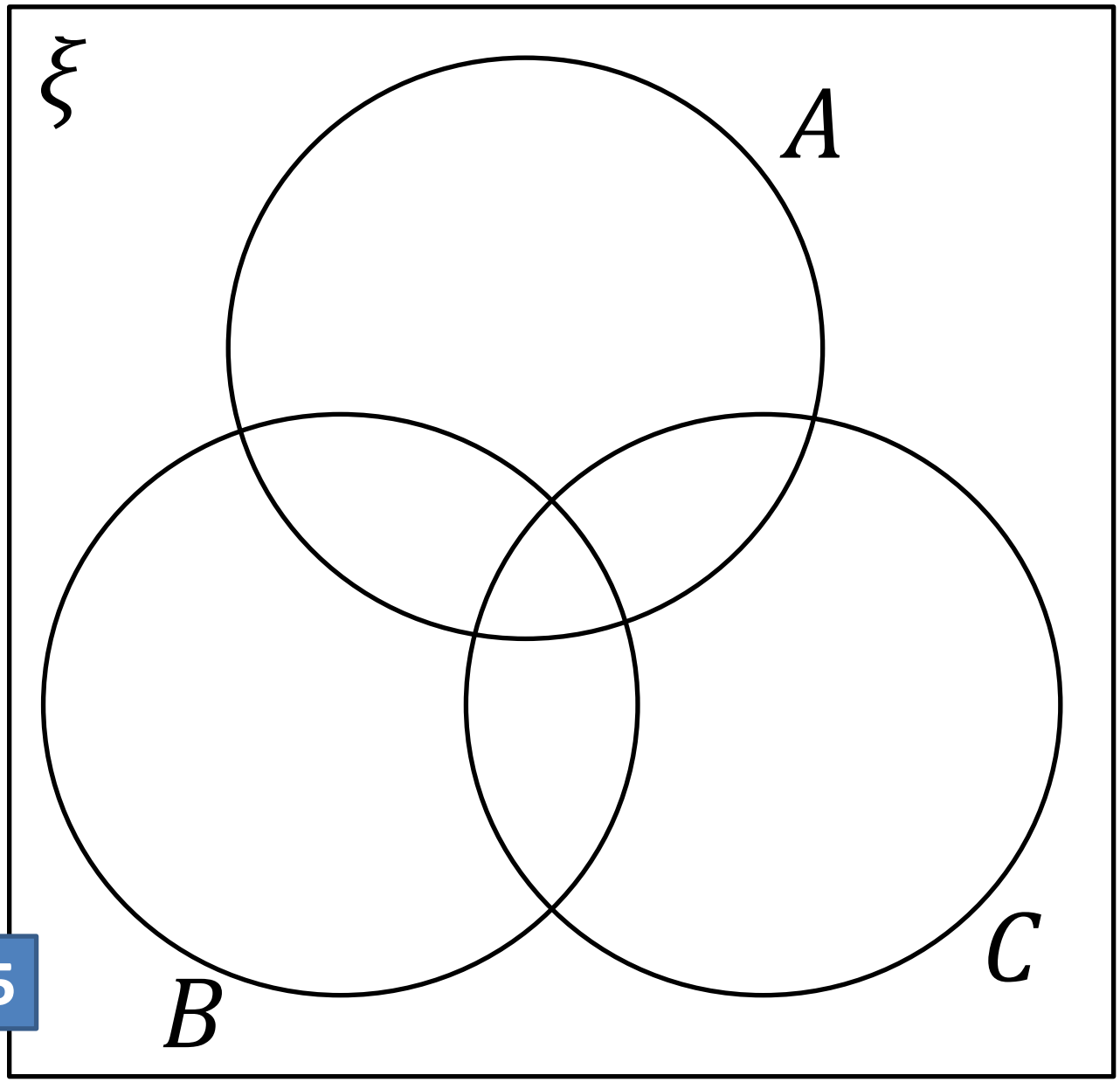
C = set of all square numbers

(Click to move!)

1 2 3

4 5 6

7 8 9 15



Test Your Understanding

ξ = whole numbers from 1 to 10

A = set of all cube numbers

B = set of all odd numbers

C = set of all multiples of 3

Bonus: If we extended ξ to include more positive integers, what's the smallest number that would appear in all three of A, B, C?

?

? Venn Diagram

Further Examples

1

$\xi = \{1,2,3,4,5,6\}$
$A = \{2,3,4\}$
$B = \{4,5\}$

Construct a Venn Diagram to show these sets.

?

2

$\xi = \{1,2,3,4,5,6\}$
$A = \{1,2,3\}$
$B = \{1\}$

Construct a Venn Diagram to show these sets.

?

Any number in B is also in A. It would therefore be a good idea to draw B inside A to show this relationship.

Elements of Sets and Subsets

An '**element**' or '**member**' is an item in a set.



$2 \in \{1,2\}$ means that 2 is a member of the set {1, 2}, i.e. it belongs to it.

$3 \notin \{1,2\}$ means that 3 is **not** a member of the set.

Note: $\in$ is a special form of the Greek letter epsilon. It just means "is a member of".

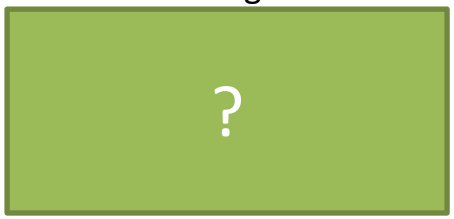
$A \subset B$ means that set A "is a subset of" B.
It means anything in A must also be in B (and $A \neq B$)

True or false?

$4 \in \{1,3,4,5\}$	False	True
$2 \notin P$ (where P is the set of all prime numbers)	False	True
$\{2,3,4\} \subset \{3,4,5\}$	False	True
$\{0,2,3\} \subset \{-1,0,1,2,3,4\}$	False	True

Exercise 1

1 $\xi = \{4,5,6,7,8\}, \quad A = \{5,6,7\}$
$B = \{6,7,8\}$
Construct a Venn Diagram to show these sets.



2 $\xi = \{1,2,3,4,5,6,7,8\}$
$A = \{1,2,3,8\}, B = \{3,4,5,8\}$
$C = \{1,5,6,8\}$
Construct a Venn Diagram to show these sets.



3 $\xi = \{1,2, \dots, 10\}$
$A =$ set of all primes
$B =$ triangular numbers
$C =$ 1 less than multiple of 4
Construct a Venn Diagram for these sets.



4 You have two sets A and B and $B \subset A$. Draw a Venn Diagram (without any numbers) that indicates the relationship between the sets.



5 You have three sets A, B and C and $A \subset C$. Draw a Venn Diagram (without any numbers) that indicates the relationship between the sets.



The **power set** of a set is the set of all possible subsets, including the empty set and itself. E.g.

$$P(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$$

a) Determine $P(\{1,2,3\})$

$=$ 

b) Determine how many members $P(A)$ has for a set of A of size n.



#2 :: Venn Diagrams involving Frequencies

Sometimes we just have one number in each region, **representing the number of items in that region**, rather than the items themselves.

Tip: The trick is to start from the centre of the diagram and work outwards.



a) ?

b) ?

c) ?

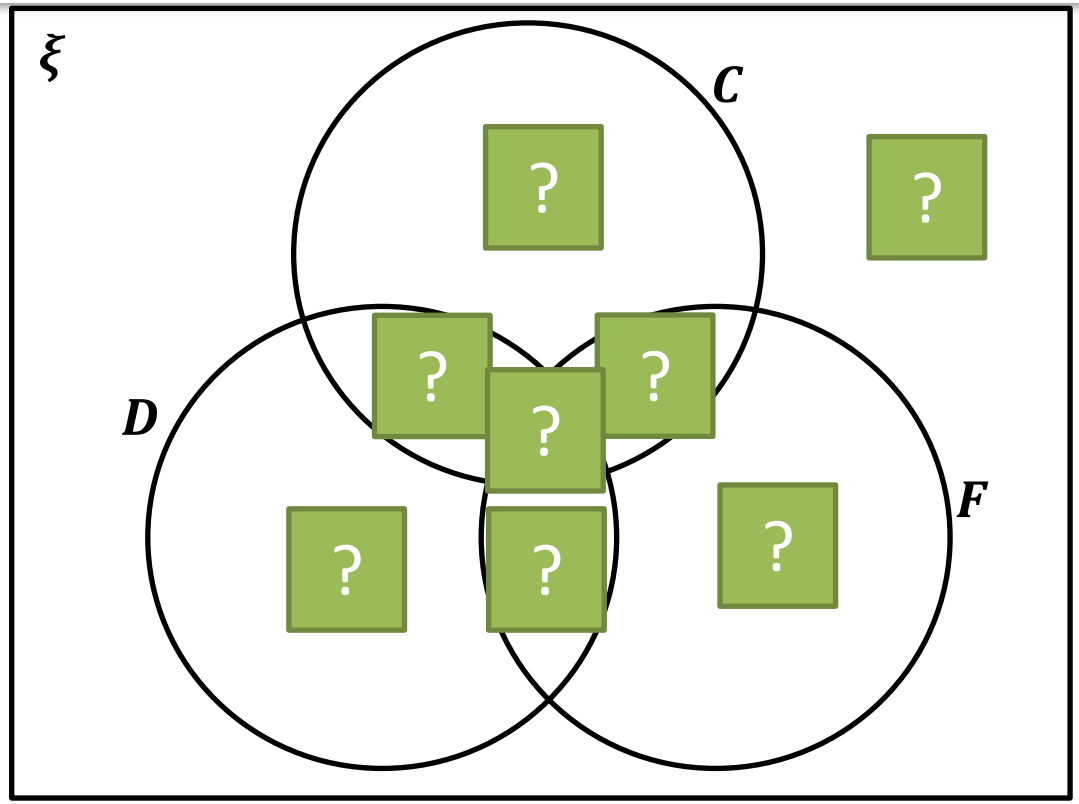
A vet surveys 100 of her clients. She finds that
25 own dogs, 15 own dogs and cats, 11 own dogs and tropical fish, 53 own cats, 10 own cats and tropical fish, 7 own dogs, cats and tropical fish, 40 own tropical fish.

Fill in this Venn Diagram, and hence answer the following questions:

a) $P(owns\ dog\ only)$

b) $P(does\ not\ own\ tropical\ fish)$

c) $P(does\ not\ own\ dogs,\ cats, or\ tropical\ fish)$

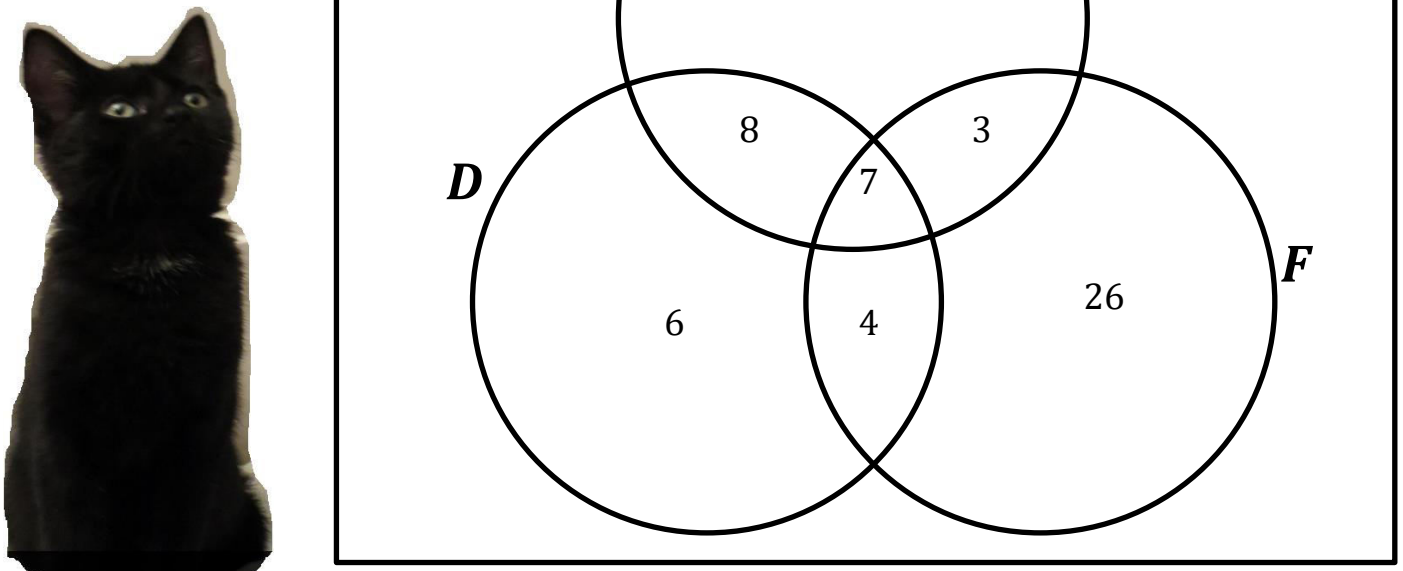


#2 :: Venn Diagrams involving Frequencies

Conditional Probabilities

Given that a randomly chosen person owns a cat, what's the probability they own a dog?

?



Test Your Understanding

Edexcel S1 Jan 2012 Q6

The following shows the results of a survey on the types of exercise taken by a group of 100 people.

65 run	48 swim
60 cycle	40 run and swim
30 swim and cycle	35 run and cycle
25 do all three	

(*a*) Draw a Venn Diagram to represent these data. **(4)**

Find the probability that a randomly selected person from the survey

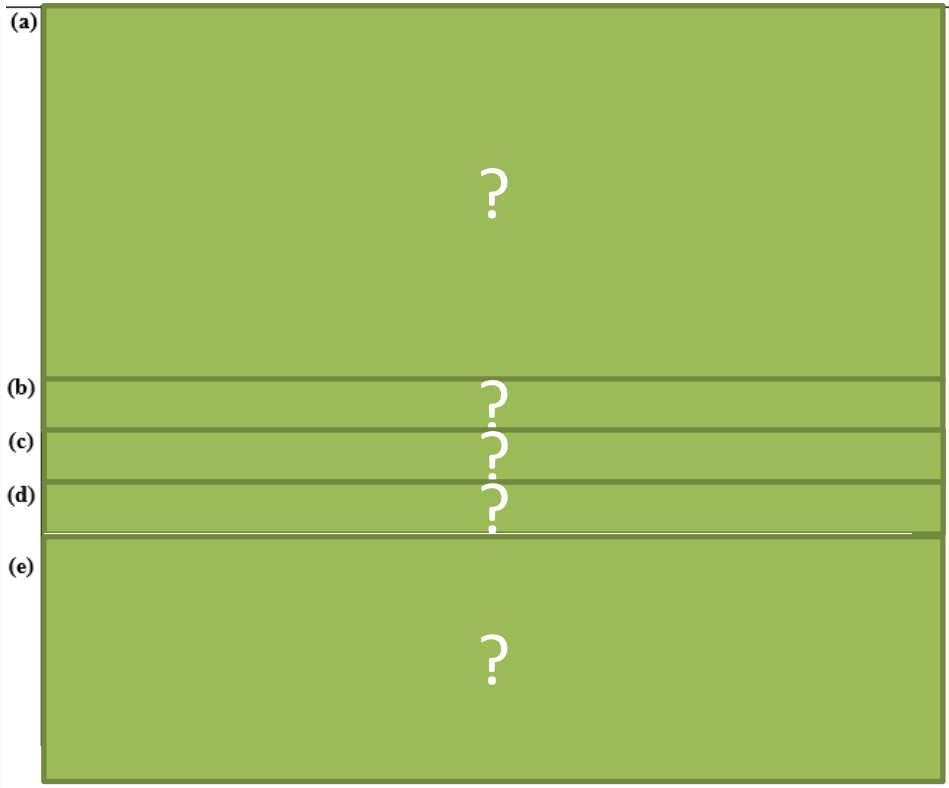
(*b*) takes none of these types of exercise, **(2)**

(*c*) swims but does not run, **(2)**

(d) takes at least two of these types of exercise. **(2)**

Jason is one of the above group. Given that Jason runs,

(*e*) find the probability that he swims but does not cycle. **(3)**



Exercise 2

(on provided sheet)

Question 1

[JMC 2002 Q11] The Pythagoras School of Music has 100 students. Of these, 60 are in the band and 20 are in the orchestra. Given that 12 students are in both the band and the orchestra, how many are in neither the band nor the orchestra?

?

Exercise 2

(on provided sheet)

2

[May 2010 Q4] The Venn diagram in Figure 1 shows the number of students in a class who read any of 3 popular magazines A, B and C.

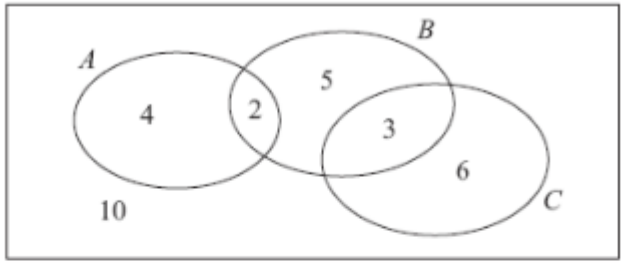


Figure 1

One of these students is selected at random.

(a) Show that the probability that the student reads more than one magazine is $\frac{1}{6}$. **(2)**

(b) Find the probability that the student reads A or B (or both). **(2)**

(c) Write down the probability that the student reads both A and C. **(1)**

Given that the student reads at least one of the magazines,

(d) find the probability that the student reads C. **(2)**

(e) Determine whether or not reading magazine B and reading magazine C are statistically independent. **(3)**

(a) ?

(b) ?

(c) ?

(d) ?

Exercise 2

(on provided sheet)

3 **[Jan 2010 Q4]** There are 180 students at a college following a general course in computing. Students on this course can choose to take up to three extra options.

112 take systems support,
70 take developing software,
81 take networking,
35 take developing software and systems support,
28 take networking and developing software,
40 take systems support and networking,
4 take all three extra options.

(*a*) Draw a Venn diagram to represent this information. **(5)**

A student from the course is chosen at random. Find the probability that the student takes

(*b*) none of the three extra options, **(1)**
(*c*) networking only. **(1)**

Students who want to become technicians take systems support and networking. Given that a randomly chosen student wants to become a technician,
(*d*) find the probability that this student takes all three extra options. **(2)**

(a) ?

(b) ?

(c) ?

(d) ?

Exercise 2

(on provided sheet)

4

[May 2008 Q5] A person's blood group is determined by whether or not it contains any of 3 substances *A*, *B* and *C*.
A doctor surveyed 300 patients' blood and produced the table below.

Blood contains	No. of Patients
only *C*	100
A and *C* but not *B*	100
only *A*	30
B and *C* but not *A*	25
only *B*	12
A, *B* and *C*	10
A and *B* but not *C*	3

(*a*) Draw a Venn diagram to represent this information. **(4)**
(*b*) Find the probability that a randomly chosen patient's blood contains substance *C*. **(2)**
Harry is one of the patients. Given that his blood contains substance *A*,

(*c*) find the probability that his blood contains all 3 substances. **(2)**
Patients whose blood contains none of these substances are called universal blood donors.

(*d*) Find the probability that a randomly chosen patient is a universal blood donor. **(2)**

(a) ?

(b) ?

(c) ?

(d) ?

Exercise 2

(on provided sheet)

5 **[Jan 2008 Q5]** The following shows the results of a wine tasting survey of 100 people.
96 like wine A, 93 like wine B,
96 like wine C, 92 like A and B,
91 like B and C, 93 like A and C,
90 like all three wines.

(a) Draw a Venn Diagram to represent these data. **(6)**

Find the probability that a randomly selected person from the survey likes

(b) none of the three wines, **(1)**
(c) wine A but not wine B, **(2)**
(d) any wine in the survey except wine C, **(2)**
(e) exactly two of the three kinds of wine. **(2)**

Given that a person from the survey likes wine A,
(f) find the probability that the person likes wine C. **(3)**

(a) ?

(b) ?

(c) ?

(d) ?

(e) ?

(f) ?

Exercise 2

(on provided sheet)

6

[May 2006 Q6] A group of 100 people produced the following information relating to three attributes. The attributes were wearing glasses, being left-handed and having dark hair.

Glasses were worn by 36 people, 28 were left-handed and 36 had dark hair. There were 17 who wore glasses and were left-handed, 19 who wore glasses and had dark hair and 15 who were left-handed and had dark hair. Only 10 people wore glasses, were left-handed and had dark hair.

(*a*) Represent these data on a Venn diagram. **(6)**

A person was selected at random from this group.
Find the probability that this person

(*b*) wore glasses but was not left-handed and did not have dark hair, **(1)**

(*c*) did not wear glasses, was not left-handed and did not have dark hair, **(1)**

(*d*) had only two of the attributes, **(2)**

(*e*) wore glasses, given they were left-handed and had dark hair. **(3)**

(a)

a ?

(b)

b ?

(c)

c ?

(d)

d ?

(e)

e ?

Exercise 2

(on provided sheet)

7 **[Jan 2005 Q5]** Articles made on a lathe are subject to three kinds of defect, *A*, *B* or *C*. A sample of 1000 articles was inspected and the following results were obtained.

31 had a type *A* defect
37 had a type *B* defect
42 had a type *C* defect
11 had both type *A* and type *B* defects
13 had both type *B* and type *C* defects
10 had both type *A* and type *C* defects
6 had all three types of defect.

(*a*) Draw a Venn diagram to represent these data. **(6)**

Find the probability that a randomly selected article from this sample had

(*b*) no defects, **(1)**

(*c*) no more than one of these defects. **(2)**

An article selected at random from this sample had only one defect.

(*d*) Find the probability that it was a type *B* defect. **(2)**

Two different articles were selected at random from this sample.

(*e*) Find the probability that both had type *B* defects. **(2)**

(a) a ?

(b) b ?

(c) c ?

(d) d ?

(e) e ?

Exercise 2

(on provided sheet)

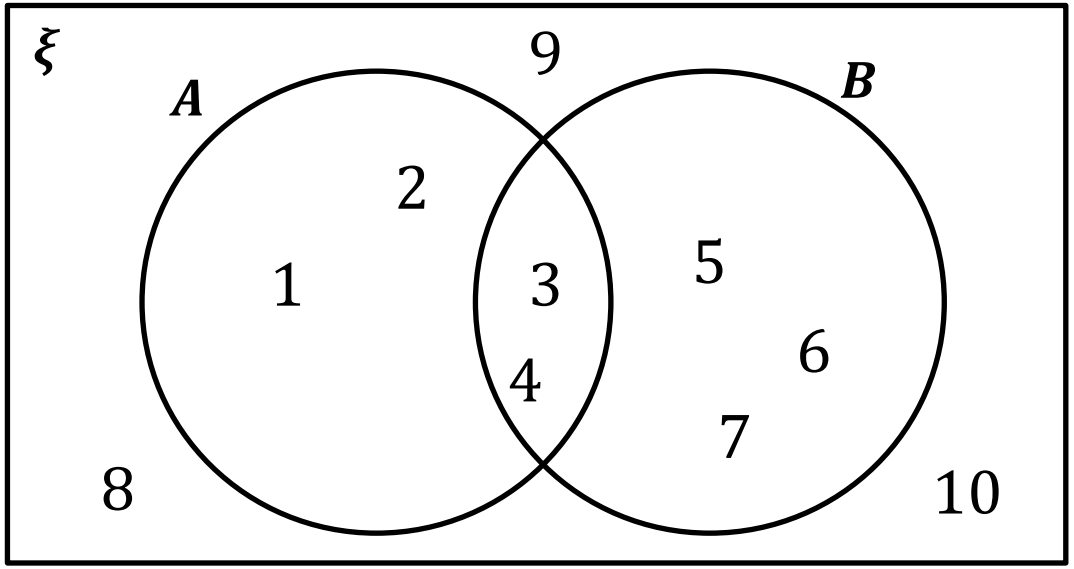
[SMC 2011 Q17] Jamie conducted a survey on the food preferences of pupils at a school and discovered that 70% of the pupils like pears, 75% like oranges, 80% like bananas and 85% like apples.
What is the smallest possible percentage of pupils who like all four of these fruits?

A. At least 10%
B. At least 15%
C. At least 20%
D. At least 25%
E. At least 70%

?

#3 Combining Sets

We have various operations on numbers, such as addition: $1 + 2 = 3$ and multiplication: $2 \times 3 = 6$
So are there similar operations on sets? Yes!



$A \cap B =$?

$A \cup B =$?

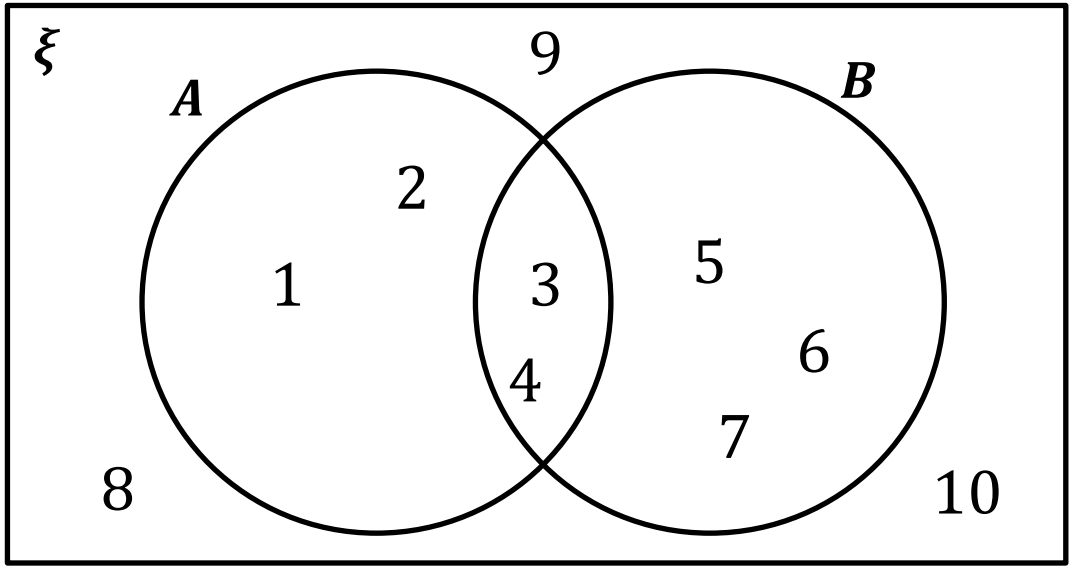
✎ $A \cap B$ is the **intersection** of A and B
It means "the things in A **and** in B"

✎ $A \cup B$ is the **union** of A and B
It means "the things in A **or** in B"*

* Things in A or B also includes things in both. Things in one but not the other is known as "exclusive or". You do not need to know this!

#3 Combining Sets

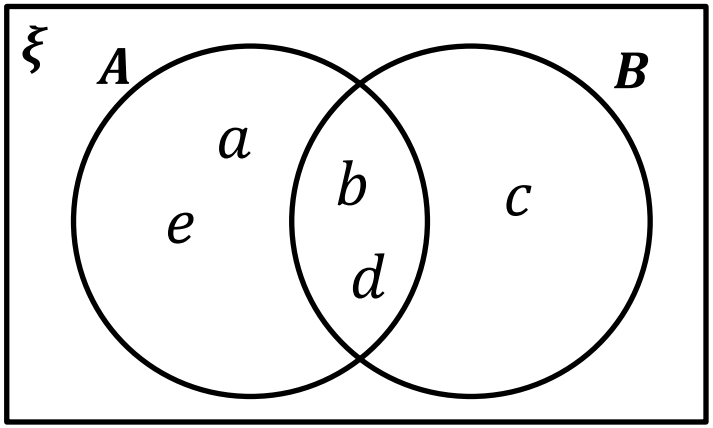
We have various operations on numbers, such as addition: $1 + 2 = 3$ and multiplication: $2 \times 3 = 6$
So are there similar operations on sets? Yes!



$A' =$?

✎ A' is the **complement** of A
It means "the things **not in A**"

Quickfire Examples



$A \cap B =$?

$A \cup B =$?

$A' =$?

$B' =$?

read this as "A and not B"

$A \cap B' =$?

$A' \cap B =$?

$A' \cap B' =$?

There is nothing that is "not in A and not in B"! ∅ is the **empty set**, as seen earlier.

$\xi = \{1,2,3, \ldots, 10\}$

$A = \{2,4,6,8,10\}$

$B = \{3,6,9\}$

$A \cup B =$?

$A \cap B =$?

$A' =$?

$A \cap B' =$?

$A' \cap B =$?

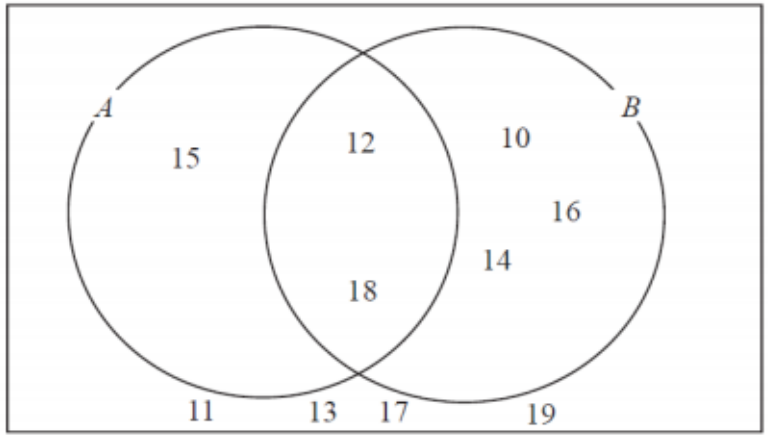
$A' \cap B' =$?

Test Your Understanding

(on provided sheet)

1 *[Edexcel Specimen Papers Set 2, Paper 2F Q26ai, Paper 2H Q5ai]*

Here is a Venn diagram.



a Write down the numbers that are in the set $A \cup B$

$A \cup B = \{$? $\}$

b *[Edexcel Specimen Papers Set 2, Paper 2F Q26b, Paper 2H Q5b]*

One of the numbers in the diagram is chosen at random.

Find the probability that the number is in set A'

?

2

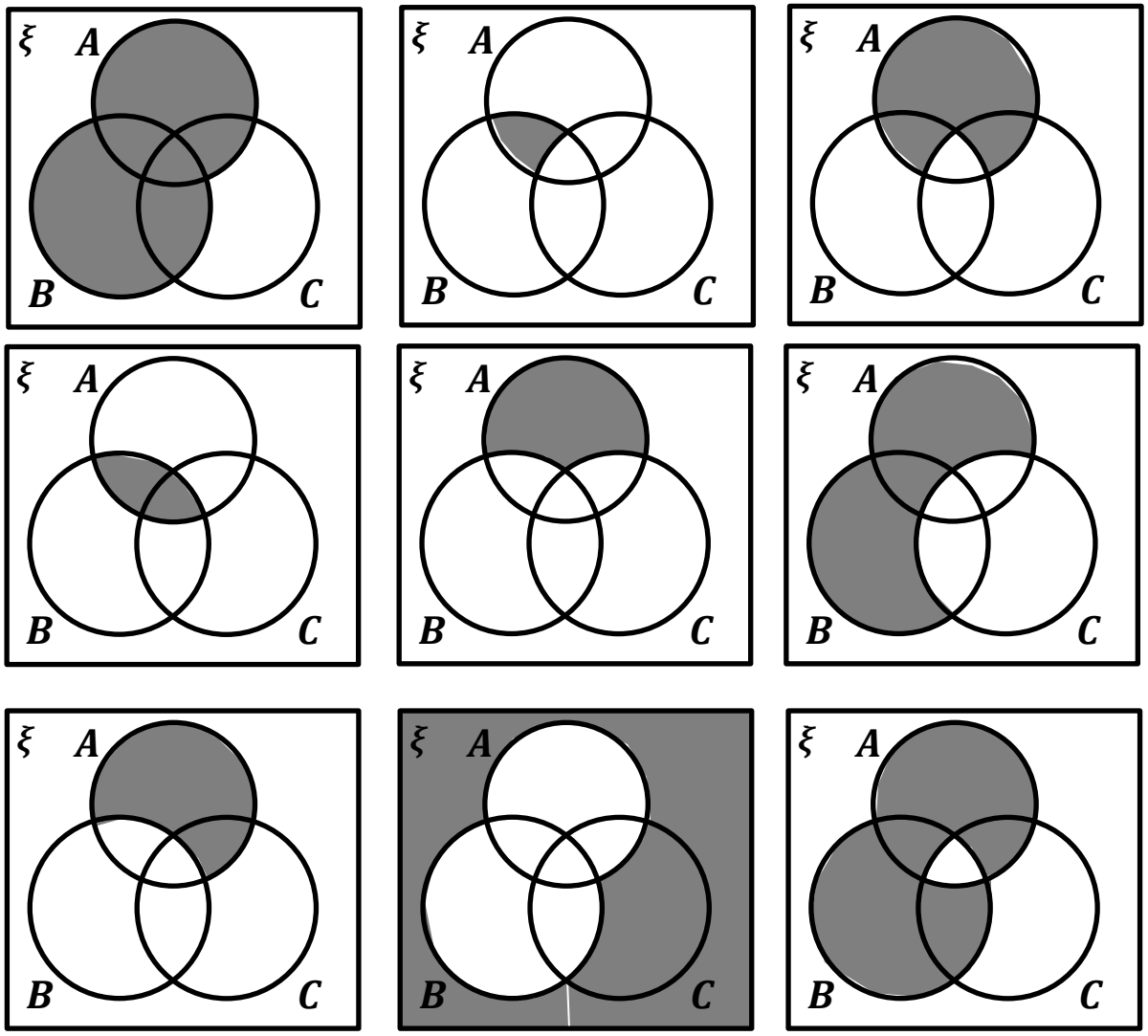
$\xi = \{$ all whole numbers $\}$
$A = \{$ factors of 60 $\}$
$B = \{$ multiples of 3 $\}$
List the members of the set $A \cap B$.

$A \cap B =$?

Matching Game!

(on provided sheet)

Match the set expressions with the indicated regions.



Click an expression below to reveal.

$A \cap B'$

$A \cap B' \cap C'$

$A \cup B$

$A' \cap B'$

$A \cap B$

$A \cap (B \cap C)'$

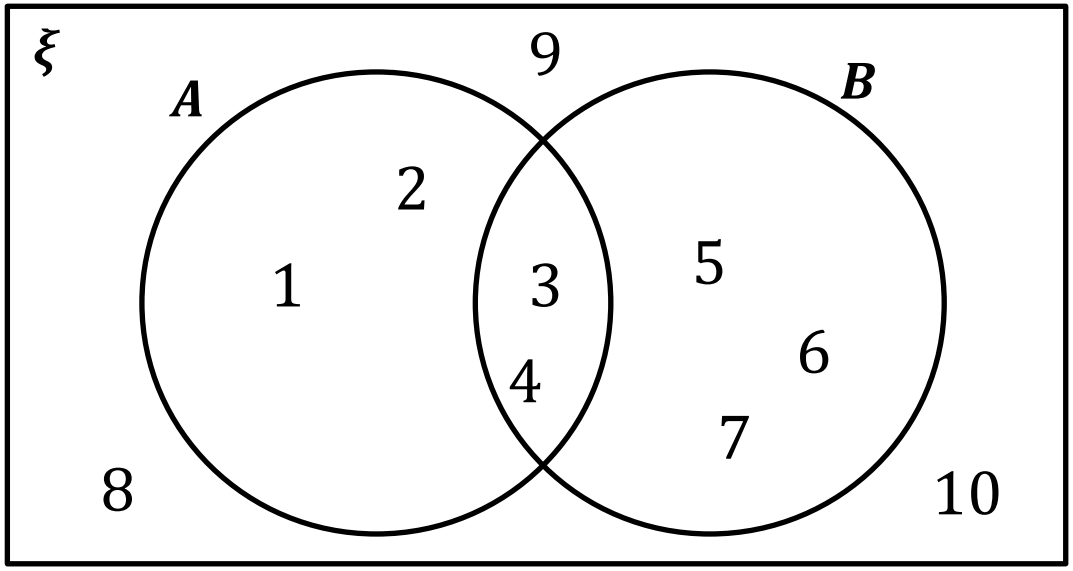
$(A \cup B) \cap (A \cap B \cap C)'$

$A \cap B \cap C'$

$(A \cup B) \cap C'$

Cardinality of Sets

The **cardinality** of a set is the size of the set.
Use $n(A)$ or $|A|$ for the cardinality/size of a set A.



$n(A) =$?

$n(B) =$?

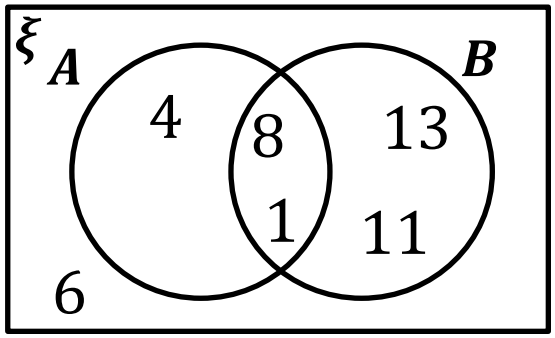
$n(A \cap B) =$?

$n(A' \cap B) =$?

Exercise 3

(on provided sheet)

1



List the numbers in:

a $A \cap B =$?

b $A \cup B =$?

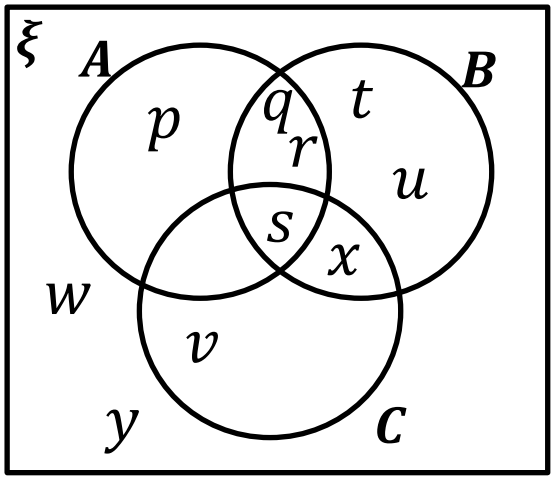
c $A' =$?

d $A' \cap B =$?

e Given that a number is chosen at random, find the probability it is in set B.

?

2



Determine the sets:

a $A \cap B =$?

b $A \cup C =$?

c $A' \cap C' =$?

d $A \cap B \cap C =$?

e $A \cap B' \cap C =$?

f $(A \cup C) \cap B' =$?

g Given that a number is chosen at random from the set $A \cup C$, find the probability it is in set B.

?

Exercise 3

(on provided sheet)

3 A and B are two sets such that:

$$n(\xi) = 20$$
$$n(A) = 14$$
$$n(A \cap B) = 3$$
$$n(B') = 12$$

Form a Venn diagram, where the number in each region is the **number** of elements in it.



4

$$A = \{f, r, o, s, t\}$$
$$B = \{b, a, r, t, o, n\}$$

Determine the sets:

a $A \cap B =$?

b $A \cup B =$?

c $A' \cap B =$?

5

$$\xi = \{1,2,3,4,5,6,7,8,9,10\}$$
$$A = \{\text{ all prime numbers }\}$$
$$B = \{\text{ all multiples of 6 }\}$$

Determine:

a $A \cup B =$?

b $A \cap B =$?

6

$$\xi = \{1,2,3,4,5,6,7,8,9,10\}$$
$$A = \{\text{ all even numbers }\}$$
$$B = \{\text{ all factors of 8 }\}$$

Determine:

a $A \cup B =$?

b $A \cap B =$?

c $A \cap B' =$?

Exercise 3

(on provided sheet)

7 Construct a Venn Diagram for sets A, B, C (without numbers) such that:

a $A \subset B, \quad B \cap C = \emptyset$

?

b $A \subset B, \quad A \cap C = \emptyset$

?

☠ A, B, C are sets such that:

$$n(A) = 20,$$
$$n(B) = 20,$$
$$n(C) = 20,$$
$$n(A \cap B) = 10$$
$$n(A \cap C) = 10$$
$$n(B \cap C) = 10$$
$$n(A \cap B \cap C) = 5$$

Determine $n(A \cup B \cup C)$.

?